Question	Answer	Mark	Mark scheme	Additional guidance
21	1 : 2	P1	for $\overrightarrow{AB} = 6\mathbf{b} - 3\mathbf{a}$ oe **or** $\overrightarrow{BA} = 3\mathbf{a} - 6\mathbf{b}$ oe **or** $\overrightarrow{OC} = 5\mathbf{a} + 5\mathbf{b}$ oe **or** $\overrightarrow{CO} = -5\mathbf{a} - 5\mathbf{b}$ oe **or** $\overrightarrow{AC} = 2\mathbf{a} + 5\mathbf{b}$ oe **or** $\overrightarrow{CA} = -2\mathbf{a} - 5\mathbf{b}$ oe	Vectors could be written on the diagram
		P1	for $\overrightarrow{AP} = -3\mathbf{a} + \frac{2}{5}(5\mathbf{a} + 5\mathbf{b})$ oe **or** $\overrightarrow{PB} = -\frac{2}{5}(5\mathbf{a} + 5\mathbf{b}) + 6\mathbf{b}$ oe	
		P1	for correct expressions, in terms of **a** and **b**, for two of $\overrightarrow{AB}$, $\overrightarrow{AP}$, $\overrightarrow{PB}$ eg two of $\overrightarrow{AB} = 6\mathbf{b} - 3\mathbf{a}$ oe, $\overrightarrow{AP} = -3\mathbf{a} + \frac{2}{5}(5\mathbf{a} + 5\mathbf{b})$ oe, $\overrightarrow{PB} = -\frac{2}{5}(5\mathbf{a} + 5\mathbf{b}) + 6\mathbf{b}$ oe	
		P1	for correct simplified expressions, in terms of **a** and **b**, for two of $\overrightarrow{AB}$, $\overrightarrow{AP}$, $\overrightarrow{PB}$ eg two of $\overrightarrow{AB} = 6\mathbf{b} - 3\mathbf{a}$ oe, $\overrightarrow{AP} = 2\mathbf{b} - \mathbf{a}$ oe, $\overrightarrow{PB} = 4\mathbf{b} - 2\mathbf{a}$ oe	
		A1	for 1 : 2 oe ratio	